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RESEARCH ARTICLE

Fairness-Oriented Charging Station Location Optimization Driven by Deep Reinforcement Learning

SIYUE YUAN[1], JIAYING FU[2], AND XIAOYIN MA[1]
[1]School of Artificial Intelligence, Beijing University of Posts and Telecommunications, Beijing 100876, China
[2]School of Remote Sensing Information Engineering, Wuhan University, Wuhan, Hubei 430072, China

Corresponding author: Siyue Yuan (2022212026@bupt.cn)

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ABSTRACT Against the backdrop of growing global attention to sustainable development and environmental protection, electric vehicles (EVs) have become an important choice for green transportation due to their zero-emission advantage. In recent years, the number of EVs in Beijing has rapidly increased, leading to an urgent demand for charging infrastructure. However, the existing layout of charging stations is imbalanced, with an over-concentration of stations in central urban areas and insufficient coverage in peripheral regions, hindering the further promotion of electric vehicles. Existing studies mostly rely on static data and traditional optimization methods, which are unable to adapt to the dynamic changes in the city, necessitating innovative strategies to improve both fairness and efficiency in location planning. This study aims to optimize the layout of charging stations in Beijing, balancing overall coverage with regional fairness to support the green transportation system. First, multi-source data including POI, population, GDP, and traffic networks are integrated, and GIS technology is used to analyze spatial distribution characteristics. Geographic detectors are then used to quantify the weight of influencing factors. Next, a fairness-oriented maximum coverage model is constructed, and Deep Reinforcement Learning (DRL) is employed to solve the problem, evaluating its advantages over traditional methods. The results show that the charging station layout in Beijing exhibits a ''dense center, sparse periphery'' pattern, with schools, hospitals, and other public facilities being the main driving factors. The DRL approach outperforms traditional methods in terms of coverage rate, fairness, and optimization efficiency, and can balance benefits and fairness by adjusting the parameter λ This study provides theoretical and practical guidance for the planning of EV charging infrastructure in Beijing.

INDEX TERMS Electric vehicle charging stations, location optimization, fairness, deep reinforcement learning, spatial analysis.

I. INTRODUCTION

The rapid urbanization and economic growth that is occurring worldwide has led to mounting challenges for numerous cities in balancing transportation development with environmental sustainability. The transportation sector, which is heavily reliant on fossil fuels, is a significant contributor to carbon emissions and air pollution. In response to these environmental concerns, governments and industry leaders are promoting green and low-carbon mobility solutions. Electric vehicles (EVs) have emerged as a pivotal alternative due to their zero-emission characteristics, contributing to cleaner air and reduced reliance on non-renewable energy sources [1].

As the number of EVs continues to increase, the demand for electric vehicle charging stations (EVCS) has grown exponentially [2]. A meticulously planned charging infrastructure is imperative for ensuring user convenience,

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promoting the widespread adoption of EVs, and enhancing the efficiency of urban transportation networks [3]. However, the deployment of EVCS is often characterized by a lack of systematic and scientific planning, resulting in uneven regional distribution and suboptimal configurations of fast and slow charging infrastructure [4]. The present distribution, characterized by its irrationality, gives rise to charging shortages during peak hours in densely populated urban areas, while coverage in remote suburban areas remains insufficient, significantly restricting the widespread adoption and application of electric vehicles [5].

Furthermore, the process of site selection frequently lacks a thorough consideration of the surrounding environment and supporting facilities. The strategic positioning of charging stations adjacent to major traffic arteries has the potential to disrupt normal traffic flow, while others do not maintain adequate safety distances from nearby flammable and explosive facilities, posing severe safety risks [6].

In response to these challenges, researchers have explored various facility location optimization models with the objective of improving EVCS siting efficiency. A significant number of studies have integrated Geographic Information Systems (GIS) with conventional optimization algorithms, leveraging spatial data to inform decision-making processes. However, most existing studies focus primarily on optimizing economic efficiency or demand coverage, often overlooking the fairness of charging resource allocation across different regions.

This study proposes a fairness-oriented EVCS siting model that ensures equitable charging accessibility while maintaining high demand coverage. Furthermore, we employ deep reinforcement learning (DRL) as an optimization framework to efficiently solve the proposed model, taking advantage of its ability to handle complex spatial decision problems. Specifically, we integrate multi-source data, including spatial information from Points of Interest (POI) data [7], and apply GIS-based analysis to evaluate the spatial distribution of existing charging stations. By incorporating DRL into the optimization framework, we improve the solution quality and adaptability of EVCS siting decisions under different urban conditions.

The contributions of this study are as follows:

- Fairness-Oriented Optimization Model: Unlike traditional EVCS siting models that primarily focus on economic efficiency, this study introduces a fairness-oriented optimization framework, ensuring equitable access to charging facilities.
- Integration of Multi-Source Data: The research leverages POI-based spatial analysis and real time urban data, enhancing the precision of charging demand estimation.
- Deep Reinforcement Learning for EVCS Siting: The application of DRL enables adaptive decision-making in response to dynamic urban environments, overcoming the limitations of static optimization approaches.

The structure of this study is organized as follows: Section II reviews the progress of related studies. Section III introduces the research area and data sources. Sections IV and V elaborate on the research methodology, experimental procedures, and comparative analysis of multiple solution approaches, followed by a discussion of the results. Finally, Section VI presents the conclusions.

II. RELATED WORK

The optimization of electric vehicle charging station (EVCS) siting has attracted significant research attention due to its critical role in supporting EV adoption and urban sustainability [8]. Existing studies primarily focus on improving EVCS deployment through geographic information system (GIS)-based spatial analysis, heuristic optimization algorithms, and deep learning techniques [9]. However, most approaches emphasize economic efficiency or demand coverage, often overlooking the fairness of charging resource allocation [10]. This section reviews the existing literature on EVCS siting from four perspectives: charging station equipment optimization, GIS-based siting studies, heuristic algorithms, deep learning-based methods and fairness-oriented siting models and the integration with DRL approaches.

A. CHARGING STATION EQUIPMENT OPTIMIZATION

Optimizing charging station equipment is a crucial research direction for improving the utilization efficiency of charging infrastructure. Numerous scholars have systematically explored the optimization of charging pile configurations from different perspectives, aiming to enhance the intelligence and automation of charging facilities [11]. In the area of dynamic charging time prediction, Li et al. [12] proposed a charging pattern classification method based on the integration of the DWDPC algorithm and a random forest model. This approach significantly improves the accuracy of charging time predictions while effectively mitigating the issue of overcharging, thereby enhancing the safety of the charging process. To address the stochastic nature of user charging behavior, Yan et al. [13] constructed a user response characteristic model. By accurately characterizing user behavior, this model enables the optimal allocation of charging resources, thereby significantly improving the overall system efficiency.

To mitigate the operational complexity and low utilization rates of charging equipment, Li et al. [14] designed a "1-to-N" automated charging pile system. This system integrates robotic arm technology, allowing a single charging pile to simultaneously serve multiple electric vehicles, thereby improving spatiotemporal utilization efficiency and reducing the impact of clustered charging on the power grid. Experimental validation demonstrated that this system, along with its two-tier iterative charging scheduling strategy, outperforms existing solutions in terms of feasibility, fairness, and real time adaptability.

B. GIS-BASED RESEARCH ON CHARGING STATION LOCATION

GIS technology, which integrates spatial analysis and data visualization capabilities, has become an essential tool in the

planning and layout of EV charging facilities. By scientifically guiding charging station siting and rational distribution, GIS technology not only enhances the service efficiency and coverage of charging infrastructure but also plays a critical role in promoting the sustainable development of the EV industry [15].

In the study of the spatial distribution of EV charging piles, Wang et al. [16] applied overlay analysis and network analysis using GIS techniques to reveal the spatial distribution characteristics and disparities of EV charging facilities in Qingdao's West Coast New Area, laying a theoretical foundation for future optimization. Qi [17] analyzed GIS-based points of interest (POI) data to explore the correlation between charging demand and influencing factors, assigning weights to each factor to develop a demand evaluation model and propose optimization strategies for facility layout. Bian et al. [18] employed ArcGIS to segment urban functional zones and incorporated factors such as traffic congestion and signal waiting times to propose an optimization scheme for charging station siting, thereby better addressing diverse user needs.

By integrating GIS with optimization algorithms, Zhang et al. [19] developed a multi-objective model based on a particle swarm optimization algorithm, demonstrating its effectiveness in improving service coverage. Nan et al. [20] utilized GM(1,1) model and ArcGIS techniques to predict charging demand and proposed the construction of 16 new charging stations in Xiamen by 2030. These studies highlight the significance of GIS-based spatial analysis in the optimal allocation of EVCS [21], [22].

C. RESEARCH ON CHARGING STATION LOCATION BASED ON HEURISTIC ALGORITHMS

Heuristic algorithms, with their efficient search capabilities and intelligent iteration characteristics, demonstrate significant advantages in handling complex location constraints, optimizing multi-objective decision-making problems, and addressing data uncertainties. This makes heuristic algorithms an effective tool for the precise positioning and optimal layout of charging stations, providing strong support for charging station location selection [23]. Yong et al. [24] introduced relevant constraints to formulate a fitness function and proposed a hybrid frog-leaping algorithm based on adaptive step size. Simulation results indicate that this algorithm outperforms traditional genetic algorithms and unimproved hybrid frog-leaping algorithms in terms of cost and time efficiency. Li [25] combined fuzzy algorithms with genetic algorithms to construct a multi-objective mathematical model, and validated its effectiveness through case analysis, providing theoretical support for charging station location selection. Zhong [26] aimed to minimize supplier costs while maximizing user satisfaction, employing the Non-dominated Sorting Genetic Algorithm II (NSGA-II) and the TOPSIS method to solve multi-objective location problems, proposing new ideas for the development of the electric vehicle industry. Qian [27] introduced an elite genetic algorithm, constructing and solving two stochastic probability coverage models—multiple coverage and single coverage—based on the relationship between the service radius of charging stations and the probability of charging demand coverage, significantly enhancing the coverage rate of charging demand. Shen et al. [10] proposed an improved genetic algorithm based on a real case in Shanghai, which, after optimization, demonstrated good convergence and solution quality in large scale instances, providing an effective decision-making tool. Li [28] proposed an improved firefly algorithm, which significantly enhanced charging efficiency by optimizing parameters such as the movement step factor, and showed better alignment with real world results compared to traditional greedy algorithms and genetic algorithms.

D. RESEARCH ON CHARGING STATION LOCATION BASED ON DEEP LEARNING

As a core technology in the field of artificial intelligence, deep learning, with its outstanding data analysis and pattern recognition capabilities, can effectively capture complex relationships between various factors and target demands. Additionally, deep learning models possess the ability to self-update and optimize, gradually becoming an important tool for solving charging station location optimization problems [29]. Yan et al. [30] validated the effectiveness of a mathematical model combined with deep reinforcement learning methods in dynamic user behavior scenarios. Jin et al. [31] proposed a charging strategy based on deep reinforcement learning, effectively coordinating electric vehicle charging and demand response, providing new solutions for charging station operations. Addressing the limitations of traditional machine learning methods in charging station location selection, Dai et al. [32] proposed a location method based on multi-source data fusion and used the LogitBoost ensemble algorithm to construct a location model. This significantly improved the model's accuracy, recall, F1 score, and ranking accuracy, thereby enhancing the accuracy and optimization effect of location selection. Li et al. [33] proposed a dynamic pricing method based on reinforcement learning and designed a three layer investment framework model including users, pricing, and location, solved using the soft actor-critic (SAC) reinforcement learning algorithm. The research results indicate that this method can achieve more economical and scientific location decisions compared to traditional methods, providing new references for charging station location and operation. In the study of neural networks solving location optimization problems, Pourvaziri et al. [34] used deep neural networks (DNN) to predict charging waiting times. By constructing a simulation model to generate training data and utilizing the Adam optimizer along with appropriate hyperparameters for model training, DNN showed superior prediction accuracy and computational efficiency compared to traditional queuing theory methods. Sun and Li [35] proposed a multi-objective neural network algorithm (MONNA) for the multi-objective

optimization problem of EVCS in Tunis. The MONNA algorithm performed excellently in optimization results, providing scientific evidence for the deployment of EVCS in Tunis, assisting decision-makers in promoting the transition to electric mobility, renewable energy utilization, and carbon emission reduction.

In summary, in the field of electric vehicle charging infrastructure, scholars have conducted in-depth research around the core goal of improving the utilization rate of charging facilities, from charging time prediction, user response characteristic analysis, to equipment automation improvement. They have combined technologies such as geographic information systems (GIS), heuristic algorithms, and deep learning to continuously optimize charging station location and operation. These studies cover a variety of technical methods and application scenarios, achieving significant progress in location optimization theory and models, providing solid theoretical support and practical guidance for the rational configuration and efficient utilization of charging stations.

While fairness-aware location models have gained traction, most existing approaches incorporate fairness through simple regularization terms or demographic weighting schemes in classical optimization frameworks. For instance, several studies adopt variants of the MCLP model to constrain service coverage disparity [36], [37], [38], but few explicitly model the minimum coverage level across regions as a fairness objective. Moreover, deep reinforcement learning (DRL) has been applied in facility siting [39], [40], but rarely in conjunction with spatial fairness constraints.

To address these challenges, this study proposes a novel approach that integrates DRL with a fairness-enhanced maximal covering location problem to optimize modeling and solving of electric vehicle charging station location problems.

III. STUDY AREA AND DATA SOURCES

A. OVERVIEW OF THE STUDY AREA

Beijing is not only the national center for politics, culture, international exchange, and technological innovation but also plays a crucial role in the national development strategy [41]. With a permanent population of 21.858 million, the large population base and high density urban construction have jointly contributed to the continuous growth of automobile ownership. Under the advocacy of green travel concepts and policy support, the adoption of electric vehicles is rapidly advancing, making the demand for charging infrastructure particularly urgent [42]. Beijing consistently regards ecological environmental protection and sustainable development as core objectives of urban construction [43]. Therefore, the rational layout of charging stations not only helps promote the widespread application of new energy vehicles but also significantly increases the proportion of green travel in the city, thereby playing a key role in reducing carbon emissions and enhancing the urban ecological environment. Beijing's diverse urban functions, significant differences in population density, and complex traffic conditions create a rich and highly representative research scenario for the study of charging station layout, offering important exemplary significance for solving similar problems in other cities [44]. Thus, this study takes Beijing as the research object for optimizing charging station location, deeply analyzing the current spatial distribution of its charging stations, and exploring reasonable location optimization strategies. Through systematic research and quantitative analysis, this study aims to propose a scientific and practical planning framework for the construction of electric vehicle charging infrastructure in Beijing, assisting in building a convenient, efficient, and green urban travel ecosystem and promoting the city's sustainable development process. The scope of the study area is shown in Figure 1.

B. DATA SOURCES

This study integrates three types of data: geographic, social, and economic, to provide comprehensive support for the optimization of charging station locations. In terms of geographic data, the Gaode Map (Amap) platform was used to capture eight categories of points of interest (POI) data in Beijing for 2024, including residential areas, office buildings, restaurants, hospitals, shopping centers, schools, theaters, and charging stations. The attributes of these datasets include latitude and longitude, category, and name, with the charging station POI data revealing geographic distribution and density characteristics, as detailed in Table 1. Additionally, traffic network data were obtained from the OpenStreetMap (OSM) website, including road types information, which aids in analyzing the spatial relationship and accessibility between charging stations and the traffic network. Social data were sourced from the WorldPop website, providing population statistics at 100-meter and 1000-meter resolutions, reflecting population distribution and activity intensity, which can be used to assess potential service demand for charging stations in different areas. Economic data were selected from the Beijing Municipal Bureau of Statistics' GDP datasets. As a key indicator of regional economic measurement, GDP is closely related to the distribution of charging stations. Analyzing the GDP levels of different areas helps evaluate the rationality of charging station distribution. Through in-depth analysis of the aforementioned data, the study comprehensively examines the spatial distribution of charging stations, population characteristics, road conditions, and surrounding POI distribution, providing scientific guidance for the optimization of charging station locations [45].

IV. METHODOLOGY

This study constructs a rigorous and systematic research framework, primarily comprising four core stages: data preprocessing, spatial analysis, model construction and solving, and optimization strategies, as illustrated in Figure 2. The study employs spatial analysis methods such as kernel density analysis, average nearest neighbor analysis, and standard deviation ellipse to examine the spatial distribution characteristics of existing charging stations in Beijing. Subsequently, the service area is finely divided into 500-meter radius

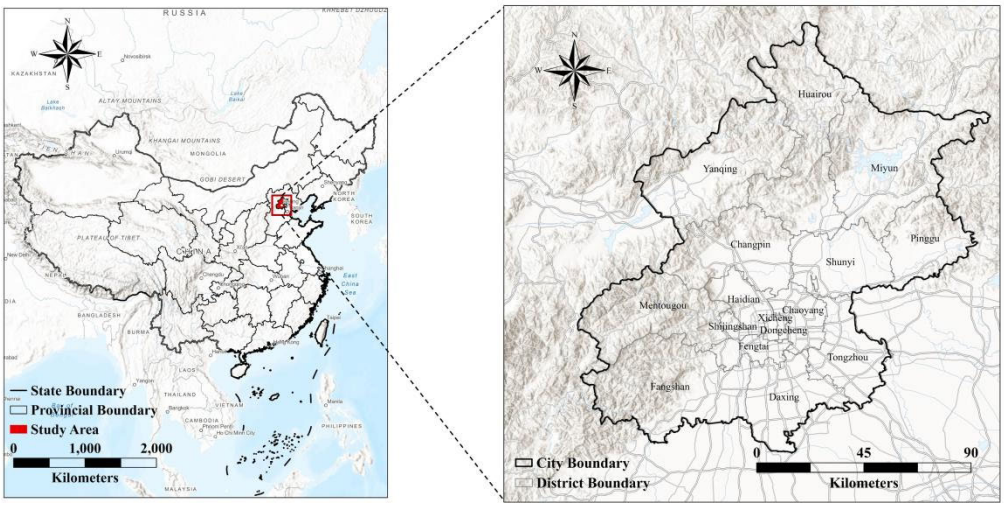


FIGURE 1. Study area: Beijing.

TABLE 1. Influencing factors.

Dimension of Impact Factor	Data Types	Code Name
Geography	Residential community points dataset	X1
	Office building points dataset	X2
	Catering points dataset	X3
	Hospital points dataset	X4
	Shopping mall points dataset	X5
	School points dataset	X6
	Theater points dataset	X7
	Road dataset	X8
Economy	GDP dataset	X9
Society	Population dataset	X10

regions, aiming to provide clear spatial units for subsequent precise analysis. Geographic detectors are used to determine the weights of influencing factors such as different types of POIs, population density, GDP, and road network density in the calculation of charging station demand intensity. Based on the determined weights, a weighted calculation of various influencing factors within the service area of each charging station is performed to accurately obtain the demand intensity of each charging station. In terms of location model construction, this study proposes a fairness-oriented Maximum Coverage Location Problem (MCLP) model, with the core objective of maximizing the total demand intensity coverage under the constraint of limited large charging stations. This model ensures optimal configuration of charging infrastructure by considering the differences in demand across different areas while ensuring fair resource allocation. To solve the model, the deep learning model SpoNet is selected, which exhibits excellent performance in large scale problems and rapid decision-making scenarios, showing high adaptability in urban facility location problems such as p-Median, p-Center, and maximum coverage issues. Ultimately, the SpoNet model is used to achieve an equitable and efficient charging station location scheme, enhancing the service capacity and coverage efficiency of urban electric vehicle charging infrastructure [39].

A. OVERVIEW OF THE STUDY AREA

Spatial analysis leverages multiple techniques to deeply explore the patterns and characteristics of geospatial data, providing scientific support for decision-making [46]. This study employs standard deviation ellipse analysis, kernel density analysis, and average nearest neighbor analysis to comprehensively dissect the spatial distribution characteristics and patterns of charging stations in Beijing. Standard deviation ellipse analysis calculates parameters such as the major axis, minor axis, and orientation angle of the ellipse, describing the spatial distribution direction and range of the study object, which is significant for analyzing spatial pattern evolution and development trends [47]. Kernel density analysis smooths spatial point data to generate a continuous density surface, visually presenting the degree of spatial aggregation and distribution trends of the study object, identifying hotspot and cold spot areas, and providing references for resource allocation and planning [48]. Average nearest neighbor analysis compares the actual distance between points and their nearest neighbors with the expected distance in a random distribution, determining the distribution pattern (clustered, random, or uniform) of the study object, aiding in the exploration of distribution patterns and underlying influencing factors [49]. The aforementioned spatial analyses are conducted using ArcGIS software, utilizing its powerful capabilities to ensure the accuracy and reliability of the analysis results, providing a scientific basis for subsequent decision-making.

B. GEODECTOR

The geographical detector is a powerful spatial analysis tool with significant advantages in uncovering the driving mechanisms behind geographic data. Its core function is to

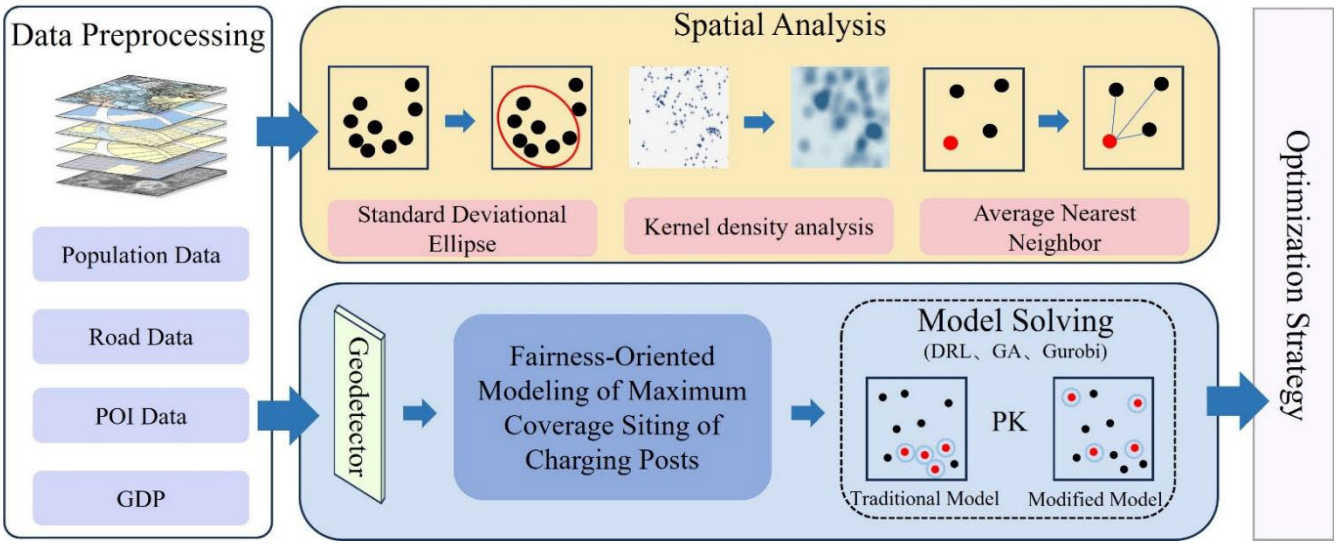


FIGURE 2. Research process framework.

identify the influence of different factors on geographic phenomena and the interaction between these factors, providing strong support for accurately understanding the causes and development patterns of geographic phenomena [50]. Using the geographical detector, the weights of various influencing factors in the calculation of charging station demand intensity were determined, enabling us to accurately obtain the demand intensity for each charging station and providing precise data support for subsequent charging station location optimization. The main tools used are the factor detector and interaction detector to explore the influence and interactive effects of different driving factors on charging stations. The factor detector is used to measure the explanatory power of a single factor on spatial distribution differences, commonly represented by the q-statistic, with its formula as follows:

$$q = 1 - \frac{\sum_{h=1}^{L} N_h \sigma_h^2}{N\sigma^2} \tag{1}$$

where q is the measure of the explanatory power of the factor on the spatial distribution of the dependent variable, with a range of [0,1]; the larger the value, the stronger the explanatory power. L is the number of subregions or categories into which factor X is divided; N_h is the number of samples in the hth subregion; σ_h^2 is the variance of the dependent variable in the hth subregion; N is the total number of samples; and σ^2 is the total variance of the dependent variable across the entire region.

The interaction detector is used to analyze whether the explanatory power of the spatial distribution is enhanced, weakened, or independent after combining two different factors.

Given two factors X_1 and X_2, the criteria for judgment are as follows:

No Interaction (Independent Effect): $q(X_1 \cap X_2) \approx q(X_1) + q(X_2)$

Enhanced Interaction (Bifactor Enhancement): $q(X_1 \cap X_2) > max[q(X_1), q(X_2)]$

Nonlinear Interaction (Strong Enhancement Effect): $q(X_1 \cap X_2) > q(X_1) + q(X_2)$

Weakened Interaction (Bifactor Suppression): $q(X_1 \cap X_2) < min[q(X_1), q(X_2)]$

C. FAIRNESS-ORIENTED MAXIMUM COVERAGE LOCATION PROBLEM FOR CHARGING STATIONS

To address both efficiency and equity in the siting of EVCS, we develop a fairness-oriented extension of the classical Maximum Coverage Location Problem. This section presents the key symbols, model formulation, and theoretical interpretation.

1) NOTATION AND PARAMETER DEFINITION

a: CANDIDATE FACILITIES AND DEMAND POINTS

I : Set of candidate locations for charging station construction.

J : Set of charging demand points, representing vehicles, residential areas, or other points requiring charging station services.

ω_j : Weight of demand j, reflecting the charging demand or service importance of that point.

Parameter a_{ij}: Defined as

$$a_{ij} = \begin{cases} 1, & \textit{if demand point i can be coverd by facility i} \\ 0, & \textit{otherwise}. \end{cases} \tag{2}$$

b: REGIONAL DIVISION AND FAIRNESS CONSIDERATION

The entire service area is divided into several subregions or groups $g \in G$ (e.g., different administrative districts, economic level areas, etc.), with each group corresponding to a set of demand points J_g.

The total demand for region g is defined as:

$$W_g = \sum_{j \in J_g} \omega_j \tag{3}$$

The coverage rate for region g is defined as:

$$C_g = \frac{1}{W_g} \sum_{j \in J_g} \omega_j y_j \tag{4}$$

c: DECISION VARIABLES

$x_i \in \{0, 1\}$: If a charging station is constructed at candidate point i,then $x_i = 1$; otherwise, $x_i = 0$.

$y_j \in \{0, 1\}$: If demand point j is covered by at least one charging station, then $y_j = 1$; otherwise, $y_j = 1$.

Auxiliary variable z: Represents the minimum coverage rate among all regions, i.e.,

$$z \leq C_g, \quad \forall g \in G$$

Facility number constraint, denoted by p, represents the maximum number of charging stations that can be constructed, constrained by:

$$\sum_{i \in I} x_i \leq p \tag{5}$$

2) MODEL FORMULATION

To balance the objectives of service efficiency and spatial equity, we propose a fairness-oriented extension of the classical Maximum Coverage Location Problem for electric vehicle charging station siting. The model incorporates a dual-objective structure by maximizing the total weighted demand coverage while simultaneously improving the coverage level of the most underserved region. The overall mathematical formulation is as follows:

$$maxmize\ \lambda \sum_{j \in J} \omega_j y_j + (1 - \lambda)\, z \tag{6}$$

$$\sum_{i \in I} x_i \leq p, \tag{7}$$

$$\sum_{i \in I} a_{ij} x_i \geq y_i,\ \forall j \in J, \tag{8}$$

$$\frac{1}{W_g} \sum_{j \in J_g} \omega_j y_j \geq z,\ \forall g \in G, \tag{9}$$

$$x_i, y_j \in \{0, 1\},\ \forall i \in I, \in J. \tag{10}$$

The model optimizes a weighted objective that combines two goals: maximizing total service coverage and improving fairness across regions. The first term $\sum_{j \in J} \omega_j y_j$ measures the total demand-weighted coverage, while the second term z represents the minimum coverage rate across all regions. The trade-off between efficiency and fairness is controlled by the parameter $\lambda \in [0, 1]$. A higher λ favors overall efficiency, while a lower value increases attention to the most disadvantaged regions. Constraint (7) ensures that the total number of charging stations deployed does not exceed the specified upper limit p. Constraint (8) enforces that a demand point j can only be considered covered if at least one selected facility is within its service range. Constraint (9) ensures that the coverage rate of each region g meets or exceeds the minimum fairness threshold z, thereby promoting interregional equity.

D. DEEP REINFORCEMENT LEARNING

To effectively solve the fairness-aware EVCS siting problem, we adopt and customize the SPO-Net framework [39], a DRL architecture originally designed for discrete spatial optimization tasks. Its encoder-decoder structure, integrated with multi-head attention and gated recurrent units (GRU), is particularly suited for capturing spatial dependencies and handling sequential decision-making in large-scale combinatorial environments.

1) MDP FORMULATION

To adapt SPO-Net to our problem, we introduce several modifications centered around fairness modeling and real-time spatial feedback. First, the EVCS siting task is formulated as a Markov Decision Process (MDP), defined as follows:

Let the siting process be represented by a tuple (S, A, P, R, γ), where:

- State (S_t): Each state S_t represents the current status of the siting decision at time step t, including the set of already selected charging stations, the spatial distribution of uncovered demand points, and the real-time coverage status of each region.
- Action (A_t): The action at each step is to select one new candidate location from the unselected station set. The selection dynamically updates the siting plan and affects subsequent demand coverage and fairness.
- Transition $P(S_{t+1}|S_t, A_t)$: The transition function models the environment's response to an action. It reflects how the newly selected location modifies the spatial coverage pattern and the distribution of remaining uncovered demand.
- Reward Function (R_t): The reward function is a weighted sum of two components: (1) total demand coverage and (2) regional fairness. Fairness is measured as the minimum coverage ratio among all spatial subregions. This dual-objective reward encourages the model to optimize both efficiency and equity.
- Discount factor γ: A scalar in [0, 1] that discount future rewards. In our setting, since siting is a finite-horizon sequential decision task, we set $\gamma = 1$.

2) SpoNet NETWORK ARCHITECTURE

The SpoNet framework, originally designed for spatial optimization problems, is adapted in this study to solve the charging station location problem efficiently. SpoNet integrates dynamic coverage information and a GRU-based decoder to enhance solution quality. The overall architecture consists of two main components, as shown in Figure 3:

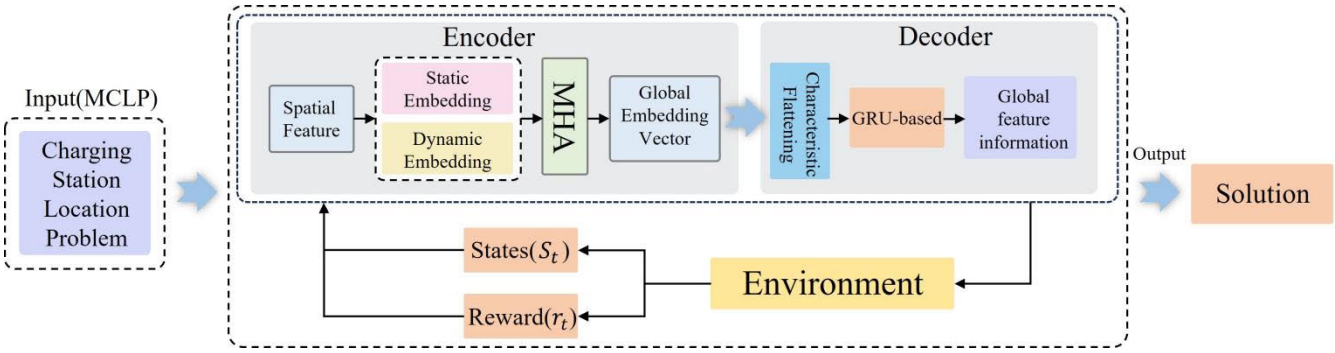


FIGURE 3. SpoNet network architecture.

a: ENCODER

The encoder transforms spatial and demand data into a meaningful representation for decision-making. It consists of two parts:

(a) Static Embedding: Captures the fixed attributes of candidate station locations, such as geographic coordinates and nearby points of interest (POI). These features are mapped into a high dimensional space through a linear projection layer.

(b) Dynamic Coverage Embedding: Introduces a real time coverage representation to track which demand points are already covered. This is updated iteratively as new locations are selected. The dynamic coverage state I_i of each node i is updated as:

$$I_i = I_i * \frac{c_i}{N_C\,(p_t)} \tag{11}$$

where p_t is the newly selected station, c_i is the rank of demand point i within its coverage set, and $N_C(p_t)$ is the total number of covered points.

The encoder uses multiple layers of Multi-Head Attention (MHA) to capture spatial dependencies, followed by a feedforward neural network with batch normalization (BN) for stability.

b: DECODER

The decoder is responsible for sequentially selecting station locations based on the encoded spatial information. It follows a recurrent structure and consists of:

(a) GRU-based Query Generator: A gated recurrent unit (GRU) is used to capture long-term dependencies in the decision process. It processes the current state and outputs a query vector for selecting the next station. The query at time t is given by:

$$q_t = GRU\,(h_{t-1}, x_t) \tag{12}$$

where h_{t-1} is the previous hidden state and x_t is the input feature of the candidate station.

(b) Attention Mechanism for Node Selection: The model applies an attention-based scoring mechanism to determine the probability of selecting each candidate site. The probability of selecting node iii is computed as:

$$p_i = \frac{e^{q_t^T k_i}}{\sum_j e^{q_t^T k_j}} \tag{13}$$

where k_i is the key embedding of node i, and softmax is used to normalize the probabilities. Overall, the extended SPO-Net effectively integrates fairness constraints into spatial decision-making through a dynamic state representation and a dual-objective reward function. This design enables the model to learn policies that balance efficiency and equity, making it well-suited for fairness-aware EVCS deployment.

V. EXPERIMENTS

A. ANALYSIS OF DRIVING FACTORS

This study utilizes the interaction detector and factor detector tools within the geographical detector model to conduct an in-depth analysis of the various factors influencing the spatial distribution of charging stations in Beijing [52]. The factor detection results clearly reveal significant heterogeneity in the influence of each factor (see Table 2).

TABLE 2. Factor detector results.

Independent Variable	Significance Level	q-Value	Explanatory Power Ranking
X1	0.1	0.500404	5
X2	0.01	0.356528	7
X3	0.1	0.558424	4
X4	0.1	0.584824	2
X5	0.1	0.572684	3
X6	0.1	0.747714	1
X7	0.01	0.338458	8
X8	0.01	0.036218	10
X9	0.01	0.219751	9
X10	0.01	0.402602	6

Ranked by the size of the q value, the order is as follows: Schools (X6) > Hospitals (X4) > Shopping Centers (X5) > Restaurants (X3) > Residential Areas (X1) > Population Density (X10) > Office Buildings (X2) > Theaters (X7) > GDP (X9) > Road Density (X8). The factor of Schools (X6) exhibits the strongest explanatory power, indicating that in the geographical spatial pattern of Beijing, the

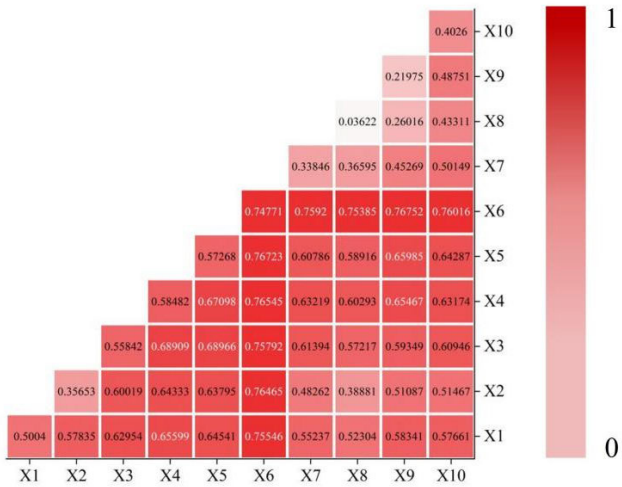


FIGURE 4. Interaction detector results.

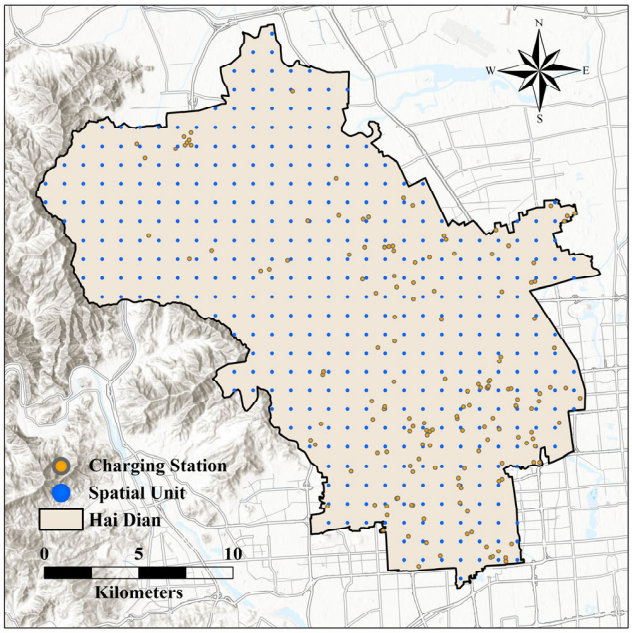


FIGURE 5. Spatial units and candidate charging station sites.

spatial distribution of charging stations is most significantly influenced by the distribution of schools. The underlying mechanism may stem from the high frequency of personnel movement and diverse travel demands around schools, which generate a strong demand for charging facilities, leading to a specific distribution pattern of charging stations near schools. Hospitals (X4) and Shopping Centers (X5), as secondary influencing factors, also play key roles in the spatial layout of charging stations. Hospitals, as public service venues with high personnel concentration, generate a large flow of people in their daily operations, including patients, medical staff, and accompanying persons, all of whom have potential demand for electric vehicle charging services, thereby significantly impacting the location and distribution of charging stations. Shopping centers, as core areas of urban commercial activity, with their strong commercial appeal and dense foot traffic, become one of the concentrated areas for electric vehicle charging demand, thus exerting a significant influence on the spatial distribution of charging stations.

The aforementioned research results not only reveal the driving factors behind the spatial distribution of charging stations in Beijing and their influence intensity but also provide a solid theoretical basis and decision support for the scientific planning, rational layout, and optimized configuration of urban charging infrastructure. This aids in better meeting the charging needs of electric vehicle users during the urban development process and promotes the sustainable development of urban transportation.

The interaction detector, as an effective tool for exploring the mechanisms influencing spatial distribution, quantifies the q values after pairwise interactions of different factors, accurately revealing the impact of factor interactions on the spatial distribution of charging stations in Beijing, as shown in Figure 4 [53].

The study found that all pairwise interactions between factors exhibit an enhanced trend, primarily manifested in a nonlinear enhancement manner. This phenomenon fully indicates that in the spatial distribution pattern of charging stations in Beijing, no single factor independently plays a decisive role; rather, the spatial differentiation pattern is determined by the synergistic and comprehensive effects of multiple factors. Among the various two factor interaction combinations, the interaction effects between Schools (X6) and GDP (X9), Shopping Centers (X5), and Hospitals (X4) are relatively significant. In stark contrast, the factor interactions between Road Density (X8) and GDP (X9), Theaters (X7), and Office Buildings (X2) are relatively weak.

B. OPTIMIZATION OF CHARGING STATION LOCATIONS

Due to equipment constraints, comprehensively solving the location optimization problem for charging stations in Beijing presents challenges. Given Haidian District's unique status and significant characteristics within Beijing, it is selected as a typical area for in-depth exploration. A grid of 1 × 1 kilometer units, totaling 433, is created as our primary spatial reference. These grid points provide precise descriptions of population, GDP, and various weighted driving factors. The study dataset also includes a total of 172 potential locations for charging stations. The coordinates of spatial units and charging stations serve as inputs to the model, aiming to carefully select 30 as central hubs for the construction of large charging stations [54]. Figure 5 shows the spatial units and candidate charging station sites.

1) ALGORITHM IMPLEMENTATION DETAILS

This section introduces the solver, programming environment, hardware configuration, and key implementation points and parameter settings of different solving algorithms (such as integer programming, heuristic algorithms, deep reinforcement learning, etc.) used in this study. Additionally,

to ensure the reproducibility of experi-mental results, we also explain the random seed settings, version management, and code open-source strategy.

a: DRL METHOD IMPLEMENTATION DETAILS

This study employs deep reinforcement learning (DRL) methods to explore solutions for the charging station location optimization problem [55]. The implementation platform is based on PyTorch, with the runtime environment being Python 3.9. The network structure proposed in SpoNet is used for training and testing, with all parameter settings consistent with that study. The network weights are initialized using a uniform distribution based on the input dimension, i.e., random sampling within the range $\left(-\frac{1}{\sqrt{d}}, \frac{1}{\sqrt{d}}\right)$, where d is the dimension of the input features. The entire model training process typically lasts for 250 epochs, with each epoch processing 250 batches, and a batchsize of 512. To ensure the convergence speed and generalization ability of the model, the Adam optimizer (Kingma and Ba 2014) is used during training, with the learning rate fixed at 10^{-4}. In terms of hardware environment, all deep reinforcement learning-related training and testing work is completed on two NVIDIA RTX 3090 GPUs and an Intel(R) Xeon(R) Gold 6230 CPU (2.10 GHz).

b: CLASSICAL METHOD IMPLEMENTATION DETAILS

Integer Programming Method: This study uses Gurobi 10.0.0 as the core mixed-integer programming solver and constructs and solves the model in the Python 3.9 environment using the gurobipy library. The constructed models include the MCLP, p-center model, and fairness-oriented model. The basic idea is to use the location status of candidate points as decision variables, and construct the objective function by combining constraints such as demand point coverage, distance limitations, and regional service balance [56].

Heuristic Method: This study uses a genetic algorithm to solve several types of problems. The algorithm implementation is based on ideas proposed in the literature, with appropriate improvements made to meet the needs of this study. The genetic algorithm uses a binary encoding scheme to represent the location status of candidate blocks, with the objective function consistent with the integer programming model, thereby achieving comprehensive optimization of charging station coverage, maximum distance, and regional fairness. Core parameter settings include a population size of 100, a maximum number of iterations of 200, a crossover rate of 0.8, and a mutation rate of 0.05. The algorithm is implemented in the Python environment using scientific computing libraries such as NumPy to perform selection, crossover, mutation, and other evolutionary operations.

c: REPRODUCIBILITY OF IMPLEMENTATION RESULTS

To ensure the reproducibility of experimental results, this study fixes the random seed in all algorithms, using methods such as random.seed() in Python and torch.manual_seed() (for DRL methods) to uniformly manage random processes. Additionally, detailed records of software versions, including Python, Gurobi, PyTorch/TensorFlow, and the operating system, are maintained.

2) EXPERIMENT DESIGN

To validate the advantages of the proposed solution methods and models for the charging station location problem, this study designs two sets of experiments. The first set of experiments focuses on comparing solution methods, aiming to verify whether the deep reinforcement learning (DRL)-based solution method can achieve or even surpass traditional methods (such as integer programming and greedy algorithms) in terms of overall coverage effectiveness and solution efficiency. In this experiment, we use a unified dataset to construct candidate charging station locations and demand points, set the same facility number constraints, and run both the DRL method and traditional solution methods on the same problem instances. During the experiment, we focus on the overall charging demand coverage rate, measured by calculating the weighted coverage rate of all demand points to assess the overall service level of the model. We also record the solution time and iteration convergence of each method and use the mean and standard deviation of results from multiple experiments to evaluate the stability and robustness of the methods. The experimental results will demonstrate the performance of the DRL method in large scale problems and under complex constraints, proving its ability to ensure high overall coverage effectiveness while significantly reducing solution time, thereby validating its competitiveness and advantages in practical applications.

The second set of experiments primarily compares the effects of model design, aiming to demonstrate that the model with fairness constraints can achieve balanced resource allocation across regions while considering overall coverage, thus better meeting the needs of practical charging station location problems. For this purpose, we construct three models under a unified dataset and regional division: first, the fairness-oriented model, with an objective function designed as $maxmize\ \lambda \sum_{j\in J} \omega_j y_j + (1-\lambda)z$, where additional fairness constraints ensure that the coverage rate of each region is not less than the auxiliary variable$z$; second, the traditional Maximum Coverage Location Problem model, which only pursues overall coverage rate; third, the p-center model, which focuses on minimizing the maximum service distance to ensure the service level of the worst region.

During the experiment, we compare the performance of the three models in terms of overall charging demand coverage rate, minimum regional coverage rate, and inter-regional coverage balance (e.g., coverage rate variance), and analyze the trade-off between overall benefit and fairness with different λ. The final experimental results will show that, while the overall coverage rate is generally satisfied, the fairness-oriented model can significantly improve the service level of the weakest region and achieve balanced distribution

of regional services, proving its clear advantages in resource balanced allocation compared to traditional MCLP and p-center models, and better meeting the actual needs of charging station location.

3) EVALUATION METRICS

To assess the performance of different solution methods and models in the charging station location problem, this study designs the following evaluation metrics from the perspectives of overall coverage effectiveness, regional balance, and solution efficiency:

Definition of Total Coverage (TC): The sum of the weighted coverage of all demand points, with the formula as follows:

$$TC = \frac{\sum_{j\in J} \omega_j y_j}{\sum_{j\in J} \omega_j} \tag{14}$$

For each subregion g, the Regional Coverage Rate (RCR) is defined as the proportion of the actual covered demand to the total demand within that region, i.e.:

$$C_g = \frac{1}{W_g} \sum_{j\in J_g} \omega_j y_j \tag{15}$$

The coverage rate of the weakest region (or minimum coverage rate) is defined as:

$$z = \min_{g\epsilon G} C_g \tag{16}$$

To measure the balance of coverage rates across regions, the variance V of regional coverage rates is used for quantification, with the calculation formula as follows:

$$V = \frac{1}{|G|} \sum_{g\in G} \left(C_g - \bar{C}\right)^2 \tag{17}$$

where $\bar{C}$ represents the mean of all regional coverage rates, calculated as:

$$\bar{C} = \frac{1}{|G|} \sum_{g\in G} C_g \tag{18}$$

Definition of Total Coverage (TC): The sum of the weighted coverage of all demand points, with the formula as follows:

$$TC = \sum_{j\in J} \omega_j y_j \tag{19}$$

4) COMPARISON RESULTS OF DEEP REINFORCEMENT LEARNING AND CLASSICAL METHODS

In this subsection, we first compare the solution results of the DRL-based method with traditional solution methods (Gurobi solver and genetic algorithm) on the fairness-oriented maximum coverage location problem for charging stations. As described in Section 3.3.2, the objective of the model in this study is to balance the maximum coverage of overall charging demand with the enhancement of service fairness across regions. The objective function is defined as:

$$maxmize\lambda \sum_{j\in J} \omega_j y_j + (1-\lambda)\, z \tag{20}$$

where λ is the weight balancing overall coverage and regional fairness, ω_j represents the weight of demand point j, y_j is an indicator variable (taking 1 if the demand point is covered, otherwise 0), and z represents the lower bound of the minimum coverage rate across regions.

To verify the solution capability of the DRL method under this objective function, we compared the performance of the DRL method, Gurobi solver, and genetic algorithm on the same dataset with $\lambda = 0.5$ fixed. Table 3 presents a comparison of different methods in terms of overall coverage rate, minimum regional coverage rate, and comprehensive objective function value, while also recording the solution time for each method. As shown in Table 3, the DRL method is on par with or slightly better than Gurobi and the genetic algorithm in terms of overall coverage rate($\frac{\sum_{j\in J} \omega_j y_j}{\sum_{j\in J} \omega_j}$), and performs better in terms of the minimum regional coverage rate z. Its objective function value is also relatively higher. Additionally, the solution time of the DRL method is significantly lower than that of traditional methods, indicating its higher solution efficiency and real time performance in large scale problems and under complex constraints.

5) COMPARISON RESULTS OF FAIRNESS-ORIENTED MODEL AND TRADITIONAL MODELS

Furthermore, to examine the trade-off effect of the objective function between overall benefit and regional fairness, we conducted a sensitivity analysis of the DRL method's performance under different λ values. Table 4 records the total coverage rate, minimum regional coverage rate, and objective function value of the DRL method, as well as the corresponding solution time, for different λ values. From the table, it can be observed that as the λ value changes, the total coverage rate and minimum regional coverage rate exhibit opposite trends: when λ is high, the model emphasizes overall coverage, achieving a higher total coverage rate; when λ is low, it focuses more on fairness, significantly improving the minimum regional coverage rate z, while the total coverage rate slightly decreases. The objective function value also shows corresponding changes with the adjustment of λ, verifying the effective regulatory role of the objective function between overall benefit and regional fairness.

Combining the data analysis from Table 3 and Table 4, the following conclusions can be drawn:(1) Under the condition of fixed $\boldsymbol{\lambda = 0.5}$, the DRL method not only matches or exceeds traditional methods in terms of total coverage rate but also excels in minimum regional coverage rate and comprehensive objective function value, while significantly reducing solution time. (2) The DRL method is quite sensitive

TABLE 3. Comparison of various metrics under different algorithms ($\lambda \mathcal{D} 0.5$).

Algorithm	Total Coverage Rate	Minimum Regional Coverage Rate	Objective Function Value	Solution Time
SpoNet	67.21%	48.18%	57.70%	0.16
Gurobi	70.67%	47.27%	58.97%	3.06
GA	63.05%	48.18%	55.62%	2.18

TABLE 4. Comparison of various metrics under different λ Values.

λ	Total Coverage Rate	Minimum Regional Coverage Rate	Objective Function Value
0	51.27%	49.09%	49.09%
0.3	64.90%	48.18%	53.20%
0.5	67.21%	48.18%	57.70%
0.8	68.59%	43.64%	63.60%
1	68.82%	43.64%	68.82%

to the λ parameter, allowing for flexible balancing between overall benefit and regional fairness by adjusting λ. This provides a basis for selecting the optimal trade-off strategy according to different demand scenarios in practical applications. Overall, the experimental results demonstrate that the DRL-based solution method has significant advantages in balancing overall charging demand coverage and regional service fairness, providing an efficient and robust solution for practical charging station location problems.

6) VISUALIZATION RESULTS

To further visually demonstrate the solution effectiveness of various methods and models, we visualized the geographic distribution of charging stations and the coverage rates of different regions. Figure 6 shows the distribution of charging station locations selected by different methods on the city map, with color and point size indicating coverage levels. It can be seen that the DRL method achieves a reasonable layout in both urban centers and edge areas, resulting in more balanced coverage.

Through the above charts and data analysis, we can visually observe that under the fairness-oriented objective function, the DRL method not only outperforms the GA algorithm in solution efficiency and stability but also has a clear advantage in balancing the maximum coverage of overall charging demand with regional service fairness.

VI. DISCUSSION

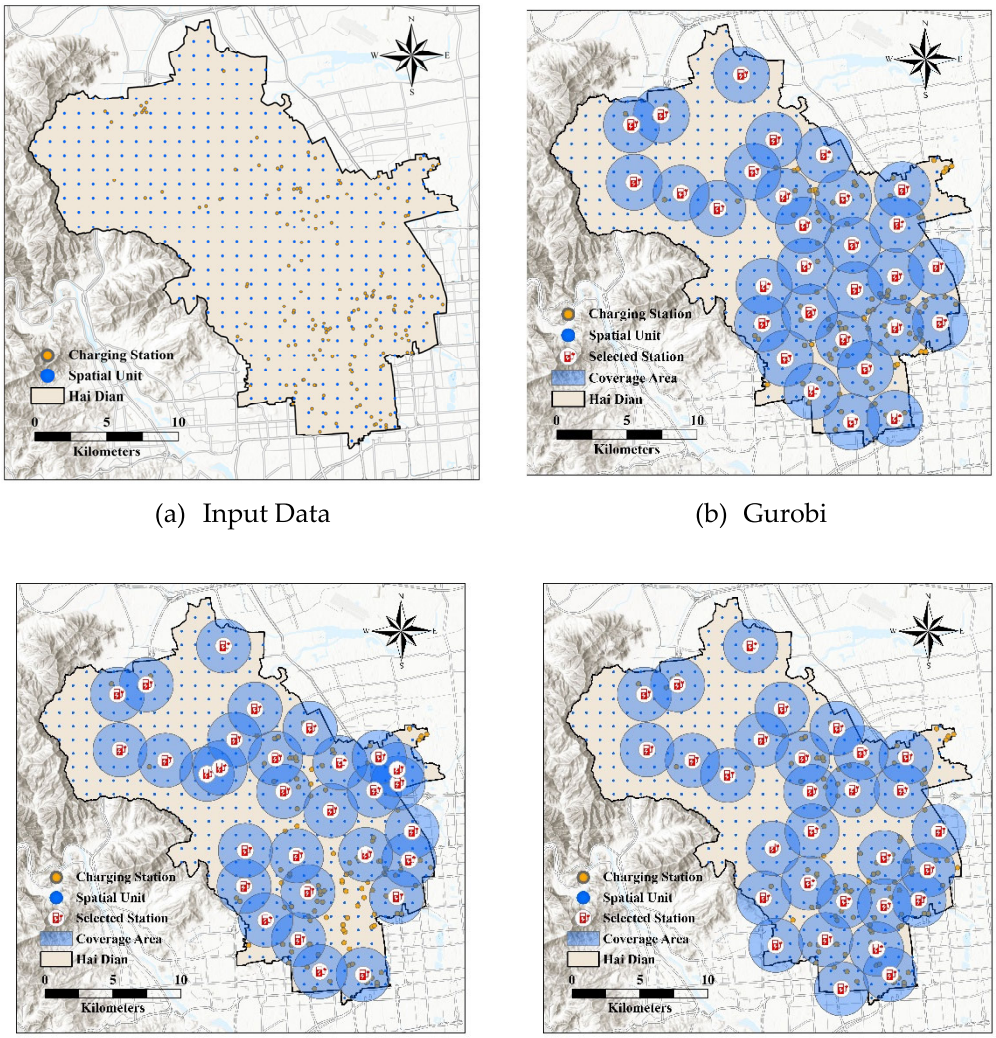
In this section, we conduct a comprehensive analysis of the experimental results to evaluate the performance of the DRL-based method in comparison with traditional solution approaches for the charging station location problem under a fairness-oriented model. The results indicate that the DRL-based method performs on par with, or even slightly better than, the Gurobi solver and genetic algorithm in terms of overall charging demand coverage rate, minimum regional coverage rate, and the value of the comprehensive objective function. In particular, the DRL approach demonstrates a significant advantage in solution time. When using the same dataset and constraints, it consistently achieves faster computation than traditional methods, highlighting its suitability for large-scale problems or those involving complex constraints. Moreover, the statistical outcomes—specifically the mean and standard deviation over multiple trials—further confirm the method's stability and robustness, providing a reliable and efficient solution for practical applications.

Furthermore, this study investigates how variations in the parameter λ influence the trade-off between maximizing overall service coverage and ensuring equitable regional distribution. The results of the sensitivity analysis indicate that the proposed fairness-oriented objective function allows for dynamic adjustment of this balance by tuning λ. Specifically, when λ takes on a higher value, the model places greater emphasis on maximizing total coverage, which can lead to insufficient service provision in peripheral or disadvantaged areas. In contrast, lower λ values markedly enhance the minimum regional coverage rate and promote a more equitable allocation of resources, albeit at the expense of a slight reduction in overall coverage. These findings underscore the effectiveness of the proposed objective function in addressing diverse planning objectives. Such a regulatory mechanism is particularly important for real-world charging station deployment, where striking an optimal balance between system-wide efficiency and spatial equity is essential to meeting the heterogeneous needs of urban and suburban regions.

The visualization results, particularly the geographic distribution maps, further validate the analysis of the experimental data. It is visually evident that the location scheme generated by the DRL method achieves a balanced and reasonable layout across both urban centers and peripheral areas, demonstrating the effectiveness of the fairness-oriented model in promoting service equity across regions. Overall, the experimental results indicate:

1. The DRL-based solution method has strong advantages in overall coverage rate, solution time, and stability, making it possible to quickly and effectively obtain optimal solutions for large scale and complex problem.

2. By introducing a fairness-constrained objective function and using the λ parameter to regulate the balance between overall benefit and regional fairness, the model not only



(a) Input Data (b) Gurobi

(c) GA (d) SpoNet

FIGURE 6. Spatial distribution of charging station layouts generated by three methods: (b) Gurobi solver, (c) GA, and (d) SPO-Net (DRL-based). Red markers indicate selected stations, and blue circles represent coverage areas.

ensures a high coverage rate of overall charging demand but also significantly improves the service level of the weakest regions, achieving balanced resource allocation.

These conclusions provide strong support for practical charging station location decisions and also demonstrate the potential application of deep reinforcement learning and fairness-oriented models in solving urban infrastructure planning problems. Future work could further consider factors such as data uncertainty, multi-objective optimization, and dynamic planning to enhance the robustness and adaptability of the model in practical applications.

VII. CONCLUSION

This study, using Beijing as a case, addresses the optimization problem of electric vehicle charging station locations by proposing a fairness-oriented Maximum Coverage Location Problem (MCLP) model. The model is efficiently solved using deep reinforcement learning (DRL) technology combined with multi-source data. The research reveals the "dense center, sparse periphery" characteristic of Beijing's charging station distribution through spatial analysis and quantifies the driving effects of factors such as schools, hospitals, and shopping centers on charging demand using geographical detectors. Experimental results indicate that the DRL method outperforms traditional methods (such as Gurobi and genetic algorithms) in terms of overall coverage rate, minimum regional coverage rate, and solution efficiency. By adjusting the parameter λ, the model can flexibly balance overall benefit and regional fairness, thereby achieving an intrinsic integration of fairness orientation and geographic spatial justice. Visualization analysis further validates the balance and rationality of the location scheme, providing directions for improvement in addressing service deficiencies in urban edge areas. This study not only enhances the scientific and sustainable layout of charging stations but also offers a replicable theoretical framework and practical

guidance for infrastructure planning in complex urban environments. Future work could further integrate dynamic traffic flow and multi-objective optimization to enhance the model's adaptability and application value, ultimately aiding in the achievement of Beijing's green transportation and sustainable development goals.

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SIYUE YUAN is currently pursuing the B.S. degree in information engineering with the School of Artificial Intelligence, Beijing University of Posts and Telecommunications, Beijing, China.

She has participated in research projects involving software development at the Engineering Research and Development Department, China Geo-Explorer (CGEO), the construction of large language models for sentiment analysis and sarcasm detection in video comments, and the development of an air–space collaborative intelligent border surveillance solution based on the integration of satellite data, UAV edge computing, and rapid ground interpretation methodologies. Her research interests include remote sensing image interpretation, the development and application of deep reinforcement learning algorithms, and spatial optimization of urban infrastructure.

Ms. Yuan was awarded the National Second Prize in the 6th Global Campus Artificial Intelligence Algorithm Elite Competition.

JIAYING FU is currently pursuing the B.S. degree in remote sensing science and technology with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China.

She has participated in research projects involving intelligent 3-D mapping of urban buildings, including the theoretical design of accuracy evaluation index systems for AI-based mapping, as well as the virtual simulation of robotic health service systems. Her research interests include remote sensing image interpretation, intelligent mapping, and urban planning.

Ms. Fu received the Gold Award in the Creativity Group Finals of the "Youth Red Dream-Building Journey" track at the China International College Students' Innovation Competition in 2024, the Silver Award in the Hubei Provincial Main Track of the same competition, the National First Prize in the 17th International Advanced Robotics and Simulation Technology Competition, and the Third Prize in the 16th "Huazhong Cup" College Student Mathematical Modeling Challenge.

XIAOYIN MA is currently pursuing the B.S. degree in information engineering with the School of Artificial Intelligence, Beijing University of Posts and Telecommunications, Beijing, China.

She has been involved in research on super-resolution reconstruction of Martian images acquired by the Moderate Resolution Imaging Camera (MoRIC) from China's Tianwen-1 mission, panoramic stitching techniques for multi-source images collected by the Zhurong rover's Navigation and Terrain Camera (NaTeCam) and the orbiter's High Resolution Imaging Camera (HiRIC) using feature-combined projection methodologies, and deep image matching (DIM) models for localization applications. Her research interests include attention mechanisms in transformer encoders, transformer-based methods for compensating missing global contextual information in convolutional neural network features, and class imbalance in classification tasks.

Ms. Ma received the Second Prize in the 2024 Beijing Undergraduate Electronics Design Contest and the National Third Prize in the 6th Global Campus Artificial Intelligence Algorithm Elite Competition.